Practical 5. Foundations of attention

JXH-1104 Neuroscience of Sport Performance

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Introduction

The associated lecture introduced the spatial cueing task and the cueing effect: slower responses when a cue directs attention to the wrong location than when it directs attention to the correct location. It also introduced the t value as a means of judging whether a difference between two conditions is larger than would be expected by chance. In this practical, each participant measures their own cueing effect, and the class data are used to compute a t value.

Aim

To measure the spatial cueing effect in each participant, and to determine whether the class's mean cueing effect is statistically significant.

Materials

- A computer with a keyboard
- LINK 1 below, to the spatial cueing task (PsyToolkit)
- LINK 2 below, to the form in which results are recorded
- LINK 3 below, to the class data workbook (view only)

TASK 1 · The spatial cueing task

1. Open the task: LINK 1, and click when ready to start.
2. Read the instructions on screen, then press the spacebar to start.
3. Respond to the location of the target as quickly and as accurately as possible: **A** for left, **L** for right. Keep the eyes on the centre of the screen throughout.
4. The task consists of 100 trials and takes a few minutes. Be patient!
5. At the end, the summary reports the response time for **valid cue conditions** and for **invalid cue conditions**. Note both values (ms).

TASK 2 · Submission of results

Open the form at LINK 2 and enter the two values noted in Task 1.

TASK 3 · Your individual cueing effect

Calculate your own cueing effect:

cueing effect = mean reaction time on invalid trials − mean reaction time on valid trials

A positive value indicates that responses were slower when the cue directed attention to the wrong location.

TASK 4 · Analysis of the whole class data

Once the class results have been submitted, they are made available in a workbook.

Follow through the analyses as I carry them out on the big screen. Each student works on their own local copy.

Step 0. Obtaining a copy. Open the class workbook at LINK 3, and save a personal copy as in Week 3 (File → Create a copy → Create a copy online). All work is carried out in the copy.

Step 1. The cueing effect of each participant. Add a column, *Difference*, containing for each participant the response time in the invalid condition *minus* the response time in the valid condition. A positive value indicates slower responses on invalid trials.

Step 2. The t value. The aim is to determine whether the mean cueing effect is larger than would be expected by chance. The t value requires the mean (M) and the standard error (SE) of the difference scores:

t = M / SE

The standard error, in turn, requires the standard deviation (SD) and the number of participants (N):

SE = SD / √N

Each quantity is computed with the corresponding Excel function, applied to the *Difference* column:

Quantity	Excel function
N	`=COUNT(range)`
M	`=AVERAGE(range)`
SD	`=STDEV.S(range)`
SE	SD divided by `SQRT(N)`
t	M divided by SE

Step 3. The p value. Excel computes p directly from the two columns of response times. For a two-tailed test on paired data:

```
=T.TEST(invalid range, valid range, 2, 1)
```

The difference is statistically significant if $p < .05$.

Step 4. Presentation. Create a column chart of the mean response time in each condition (y-axis beginning at zero), and a column chart of the difference scores sorted from smallest to largest, in which each bar represents one participant.

Questions. Is the mean cueing effect positive? How does the individual cueing effect from Task 3 compare with the class mean? Does the t value lie outside the range of approximately -2 to $+2$, and is p smaller than .05? How many participants show a negative cueing effect?

Additional exercises

- How does the t value change if using very few participants, for example, only five? Explain why. (Hint: the t value is a ratio of the mean to the standard error, and the standard error depends on the number of participants.). Exercise R4 below tests this prediction on simulated data.
- Sebanz and Shiffrar (2009) reported that experts classified passes "above chance". Describe how the same logic would be applied to a comparison between a mean accuracy and 50%.

Additional exercises in R (optional)

These exercises are optional and any associated knowledge is not assessed. They repeat the analysis of this practical in **R**.

To run a cell, select **Run Code**. The code in each cell can be edited and run again. The data are simulated/made-up: they represent ten fictitious participants, not the actual class.

R1 · Entering data

Each value in `valid` and `invalid` is one participant's mean reaction time (ms). The cueing effect of each participant is calculated by subtraction.

```
valid   <- c(312, 345, 298, 367, 330, 355, 301, 342, 318, 336)
invalid <- c(334, 337, 309, 402, 325, 372, 322, 349, 341, 350)

cueing <- invalid - valid
cueing
```

R2 · The t value, calculated from the formula

The same quantities as in the class workbook: number of participants, mean, standard deviation, standard error, and t. Run R1 first.

```
n  <- length(cueing)
mean  <- mean(cueing)
stdev  <- sd(cueing)
sterr <- stdev / sqrt(n)
t_value <- mean / sterr

c(n, mean, stdev, sterr, t_value)
```

R3 · The paired t-test

R calculates the same t value in a single line, together with the p value.

```
t.test(invalid, valid, paired = TRUE)
```

Questions. Is p smaller than .05?

R4 · Fewer participants

`[1:5]` selects the first five values only.

```
t.test(invalid[1:5], valid[1:5], paired = TRUE)
```

Questions. How do t and p change? With reference to the formula in R2, which quantities are responsible for the change?

R5 · The distribution of cueing effects

```
hist(cueing, breaks = 8,
     main = "Cueing effect per participant",
     xlab = "Invalid - valid (ms)")
abline(v = 0, col = "red", lwd = 3)
```

R6 · The t value when there is no true difference

As in the lecture: 10000 simulated samples of ten participants in which the true cueing effect is zero. The red line marks the t value from R2.

```
t_null <- replicate(10000, {
  x <- rnorm(10, mean = 0, sd = 13)
  mean(x) / (sd(x) / sqrt(10))
})
hist(t_null, breaks = 40, main = "t values with no true difference", xlab = "t")
abline(v = t_value, col = "red", lwd = 3)
mean(abs(t_null) >= abs(t_value))
```

Question. The last line reports the proportion of simulated samples with a t at least as extreme as the observed one. How does it compare with the p value in R3?

Expected results

- R1: cueing effects 22, −8, 11, 35, −5, 17, 21, 7, 23, 14 ms.
- R2: n = 10, mean = 13.7 ms, SD = 13.1 ms, SE = 4.1 ms, t = 3.31.
- R3: t = 3.31, df = 9, p = .009. The cueing effect was statistically significant: responses were significantly slower on invalid than on valid trials.
- R4: t = 1.36, df = 4, p = .25. Not statistically significant. The mean is similar, but the standard deviation is larger and n is smaller, so the standard error is larger and t is smaller.
- R5: most values lie to the right of zero; two participants show a negative cueing effect.
- R6: the proportion is close to the p value in R3 (it varies slightly each time the cell is run, because the samples are random).

If the task could not be completed

The following values may be used: valid trials 345 ms; invalid trials 378 ms.